

Backend DB-Backed RAGAS Agent Report

Evaluation date: 2026-06-21 | Target: backend /chat/query with DB-saved contexts | Scope: 5 Kubernetes cases

Objective

Run RAGAS through the backend API, then verify that answer sources were saved in the database. A database-checking sub-agent independently inspected the local Postgres state.

Method

- Started a database/evidence checker agent to inspect local Docker/Postgres state.
- Confirmed the normal backend pipeline URL /webhook/retrieve returned an empty body through n8n.
- Temporarily restarted backend with PIPELINE_URL=http://n8n:5678/webhook/retrieve-eval because /webhook/retrieve-eval returns JSON with answer and sources.
- Created a local eval tenant/user for tenant 22222222-2222-2222-2222-222222222222, matching the n8n evaluation corpus tenant.
- Ran backend /chat/query evaluation with --contexts-from-db so answers came from the API and contexts came from saved chat_messages.sources.
- Ran RAGAS on 5 Kubernetes evaluation cases with OpenAI as the independent evaluator.
- Restored backend to the normal compose configuration after the run.

RAGAS Summary

Metric	Score	Threshold	Status
Faithfulness	0.6381	0.85	Fail
Answer relevancy	0.9402	0.80	Pass
Context precision	0.9178	0.75	Pass
Context recall	1.0000	0.80	Pass

Quality gate failed because faithfulness was below the 0.85 threshold. Answer relevancy, context precision, and context recall passed.

Retrieval/API Summary

- Cases: 5; answerable cases: 5.
- Citation coverage: 1.0.
- Mean latency: 5041.888 ms; P95 latency: 5822.47 ms.
- All 5 cases returned backend HTTP 200.
- The evaluator read contexts from local chat_messages.sources, not directly from the API response.

Per-Case Scores

Case	Faithfulness	Answer Relevancy	Context Precision	Context Recall
K8S-EVAL-001	0.3333	0.9349	1.0000	1.0000
K8S-EVAL-002	1.0000	0.9661	0.7556	1.0000
K8S-EVAL-003	1.0000	0.9712	1.0000	1.0000
K8S-EVAL-004	0.2857	0.8602	1.0000	1.0000
K8S-EVAL-005	0.5714	0.9688	0.8333	1.0000

Database Evidence

Database checker findings before and around the run:

- Local rag-postgres was reachable and healthy.
- Local app DB had 1 original tenant and 1 original user before adding the eval tenant/user.
- Local documents: 0 rows.
- Local document_chunks: 0 rows.
- Local ephemeral_chunks: 0 rows.
- Historical chat_messages existed and had assistant sources, but those were not proof of the current evaluation corpus.

Post-run evidence for eval tenant 22222222-2222-2222-2222-222222222222:

- Local documents: 0 rows.
- Local document_chunks: 0 rows.
- Latest backend-generated assistant rows had source_count=5.
- Saved source objects included chunk_text, document_id, and score.
- First source title in latest rows: kubernetes_troubleshooting/cluster_debugging_runbook.txt.

Interpretation

Backend persistence is working because answers and returned sources were saved into chat_messages.sources. However, local app documents and document_chunks are not populated, so the retrieved evaluation corpus appears to live behind the n8n workflow configured Postgres credential rather than the local app DB.

Key Caveat

This run validates backend /chat/query, backend source persistence, RAGAS scoring over saved contexts, and the saved source shape. It does not validate that local app document_chunks powered retrieval, because local document_chunks is empty. For strict end-to-end corpus validation, align n8n Postgres with the local app DB or ingest the same evaluation corpus into the local app DB.

Artifacts

- JSON: artifacts/evaluation_results/application_suite_eval_20260621T111722Z.json
- CSV: artifacts/evaluation_results/application_suite_eval_20260621T111722Z.csv
- Markdown: artifacts/evaluation_results/application_suite_eval_20260621T111722Z.md
- HTML: artifacts/evaluation_results/application_suite_eval_20260621T111722Z.html

Recommended Next Step

Populate local documents and document_chunks for tenant 22222222-2222-2222-2222-222222222222, or repoint n8n Postgres to local rag-postgres. Then rerun:

```
python -m evaluation.run_eval --suite-dataset --base-url http://localhost:8000 --email eval-ragas@example.com
--password evalpass123 --contexts-from-db --db-url postgresql+asyncpg://raguser:changeme@localhost:5432/ragplatform
```